

Molecular and Systems-Level Effects of Matcha on Human Health: An Exploratory Bioinformatics Analysis

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Abstract

Matcha is a powdered green tea rich in biologically active compounds, including catechins, caffeine and L-theanine. Among these compounds, epigallocatechin gallate (EGCG) has attracted considerable scientific interest because of its reported antioxidant, anti-inflammatory and cardiometabolic properties. Despite numerous experimental studies, the molecular mechanisms underlying these biological effects remain incompletely understood.

This project aimed to investigate the potential molecular targets and biological pathways associated with major matcha compounds using publicly available bioinformatics resources. Chemical information was obtained from PubChem, while SwissTargetPrediction was used to identify predicted human protein targets. Functional enrichment analysis was performed using Gene Ontology (GO), KEGG and Reactome databases, followed by protein–protein interaction analysis using STRING and NetworkX.

The analysis identified approximately 96 unique predicted human protein targets for EGCG. Functional enrichment highlighted pathways associated with protein phosphorylation, extracellular matrix organisation, PI3K–Akt signalling, MAPK signalling and VEGF signalling. Protein interaction analysis revealed a highly interconnected network in which BCL2, MMP9, HIF1A, ESR1 and MTOR were identified as major hub genes.

These findings suggest that EGCG may influence multiple interconnected signalling pathways involved in cellular regulation. However, because the analyses rely primarily on computational prediction and publicly available databases, experimental validation is required before any biological or clinical conclusions can be drawn.

Keywords: Matcha, EGCG, Bioinformatics, Pathway enrichment, STRING, Protein–protein interaction network, Systems biology.

1. Introduction

Matcha is a finely powdered form of green tea produced from shade-grown leaves of *Camellia sinensis*. Unlike conventional green tea infusions, matcha is consumed as the whole leaf, resulting in higher exposure to several bioactive phytochemicals, particularly catechins. The most abundant and biologically active catechin is epigallocatechin gallate (EGCG), which has been extensively investigated because of its antioxidant, anti-inflammatory and potential metabolic effects.

In addition to EGCG, matcha contains other catechins including epigallocatechin (EGC), epicatechin gallate (ECG), epicatechin (EC) and catechin, together with caffeine, L-theanine and gallic acid. These compounds have been associated with a wide range of biological activities, including modulation of oxidative stress, inflammatory signalling, glucose metabolism, cardiovascular function and neuronal activity. Nevertheless, many reported health benefits remain under investigation, and the molecular mechanisms responsible for these effects are not yet fully understood.

Advances in computational biology enable researchers to investigate potential molecular interactions before conducting laboratory experiments. Public databases such as PubChem, SwissTargetPrediction, STRING and Enrichr provide valuable resources for identifying predicted molecular targets, biological pathways and protein interaction networks. These approaches cannot demonstrate causality, but they can generate biologically plausible hypotheses that guide future experimental research.

The objective of this project was to investigate the molecular and systems-level effects of major matcha bioactive compounds, with particular emphasis on EGCG. The study combined chemical annotation, target prediction, pathway enrichment analysis and protein–protein interaction network analysis to identify biological pathways and key regulatory proteins that may contribute to the reported biological activities of EGCG.

2. Materials and Methods

2.1 Identification of Bioactive Compounds

The first stage of the project involved identifying the principal bioactive compounds present in matcha green tea based on published scientific literature. Eight representative compounds were selected for further analysis:

- Epigallocatechin gallate (EGCG)
- Epigallocatechin (EGC)
- Epicatechin gallate (ECG)
- Epicatechin (EC)
- Catechin
- Gallic acid
- Caffeine
- L-theanine

These compounds represent the major catechins, alkaloids, amino acids and phenolic acids commonly reported in matcha.

2.2 Chemical Annotation

Chemical information for each compound was retrieved from the PubChem database using its REST API. The collected information included:

- PubChem Compound ID (CID)
- Molecular formula
- Molecular weight
- Canonical SMILES
- IUPAC name

The retrieved information was organised into a structured dataset using Python and the pandas library.

2.3 Protein Target Prediction

SwissTargetPrediction was used to identify potential human protein targets for EGCG. The canonical SMILES representation of EGCG was submitted to the web server, and the organism was specified as *Homo sapiens*.

Predicted targets were downloaded as a CSV file and imported into Python for further processing. Duplicate entries were removed to generate a list of unique predicted human genes.

2.4 Functional Enrichment Analysis

Functional enrichment analysis was performed using the GSEAPy package, which provides access to the Enrichr database.

The following annotation libraries were analysed:

- Gene Ontology Biological Process (GO BP)
- Gene Ontology Molecular Function (GO MF)
- Gene Ontology Cellular Component (GO CC)
- KEGG Pathways
- Reactome Pathways

The analysis aimed to identify biological processes and signalling pathways significantly enriched among the predicted EGCG target genes.

2.5 Protein–Protein Interaction Network

Protein interaction analysis was performed using the STRING database with the organism restricted to *Homo sapiens*.

The resulting interaction network was imported into Python and analysed using the NetworkX library.

Node degree was calculated to identify hub proteins within the interaction network.

2.6 Data Visualisation

All analyses were performed using Python.

Figures were generated using:

- Matplotlib
- NetworkX

Protein interaction files were additionally exported in a format compatible with Cytoscape for future visualisation and network exploration.

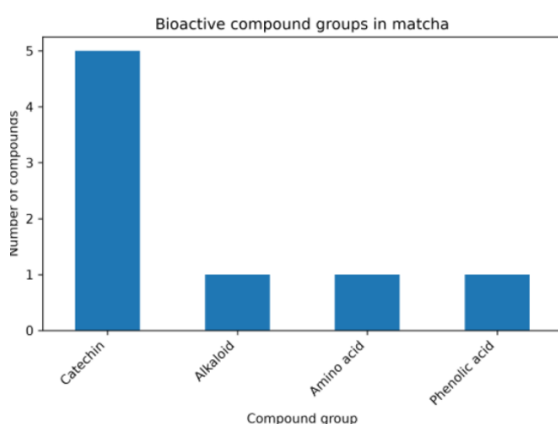
3. Results

3.1 Bioactive Compound Identification

Eight major bioactive compounds present in matcha were selected for analysis. Most of the compounds belonged to the catechin family, while caffeine represented the alkaloid group, L-theanine represented amino acids and gallic acid represented phenolic acids.

The compound distribution is shown in Figure 1.

Figure 1. Distribution of major bioactive compound groups identified in matcha.



3.2 Protein Target Prediction

SwissTargetPrediction identified approximately 100 predicted protein targets for EGCG. After removing duplicate entries, 96 unique human genes were retained for downstream analysis.

Target classification demonstrated that kinases represented the largest protein family, followed by proteases, lyases, oxidoreductases and transferases. This distribution suggests that EGCG may interact with proteins involved in diverse cellular signalling processes.

3.3 Gene Ontology Enrichment Analysis

Gene Ontology Biological Process enrichment identified several significantly enriched biological processes.

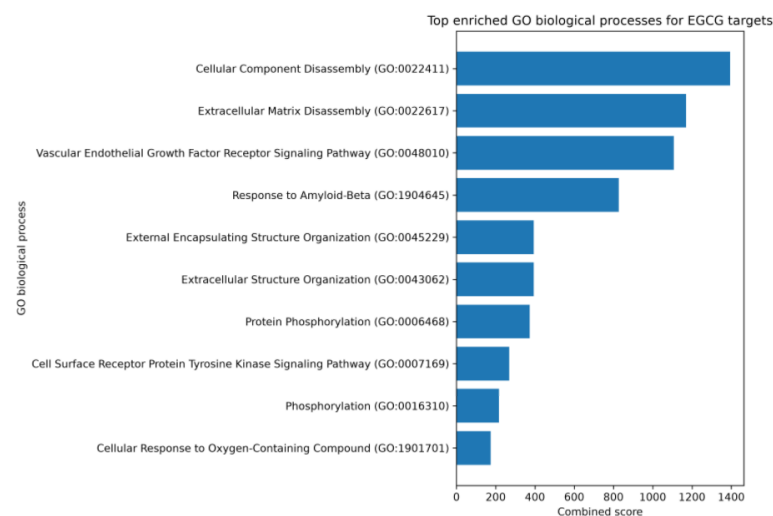
The highest-ranked biological processes included:

- cellular component disassembly
- extracellular matrix disassembly
- VEGF receptor signalling pathway
- protein phosphorylation
- extracellular structure organisation
- cellular response to oxygen-containing compounds

These results indicate that predicted EGCG targets participate primarily in signalling pathways associated with extracellular matrix remodelling, cellular communication and regulation of protein phosphorylation.

The enrichment results are presented in Figure 2.

Figure 2. Top enriched Gene Ontology Biological Processes for predicted EGCG target genes.



3.4 KEGG Pathway Enrichment

KEGG pathway analysis identified several biologically relevant signalling pathways.

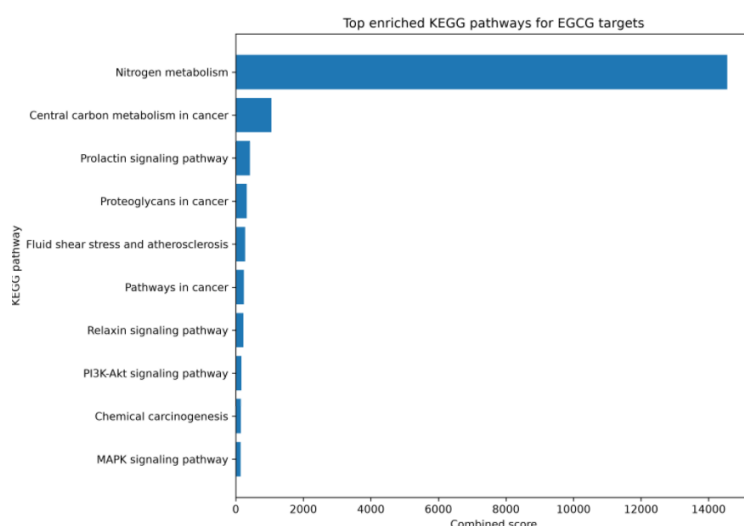
Among the enriched pathways were:

- PI3K–Akt signalling pathway
- MAPK signalling pathway
- Relaxin signalling pathway
- Fluid shear stress and atherosclerosis
- Proteoglycans in cancer
- Pathways in cancer

Although nitrogen metabolism achieved the highest combined enrichment score, pathways associated with intracellular signalling are likely to be more relevant to the known biological activities of EGCG.

The enrichment results are shown in Figure 3.

Figure 3. Top enriched KEGG pathways associated with predicted EGCG targets.



3.5 Reactome Pathway Enrichment

Reactome analysis produced similar findings.

Several significantly enriched pathways were associated with:

- PI3K/AKT signalling
- VEGF signalling
- receptor tyrosine kinase signalling
- extracellular matrix degradation
- interleukin signalling
- signal transduction

These pathways are consistent with processes involved in cellular growth, survival, inflammatory signalling and tissue remodelling.

The results are presented in Figure 4.

Figure 4. Top enriched Reactome pathways associated with predicted EGCG targets.



3.6 Protein–Protein Interaction Network

To investigate the functional relationships among predicted EGCG target proteins, a protein–protein interaction (PPI) network was constructed using the STRING database. The interaction data were imported into Python and analysed using the NetworkX package.

The resulting network consisted of approximately 96 protein nodes connected by more than 500 interactions, indicating a highly interconnected molecular network rather than isolated protein targets. Such connectivity suggests that EGCG may influence multiple biological

pathways simultaneously through interacting signalling proteins rather than acting on a single molecular target.

Degree centrality analysis was performed to identify highly connected proteins (hub genes) within the interaction network. The proteins with the highest connectivity included BCL2, MMP9, HIF1A, ESR1, SRC, GSK3B, MTOR, KDR, MMP2 and PIK3CA.

Several of these proteins are well-established regulators of apoptosis, angiogenesis, intracellular signalling and extracellular matrix remodelling. Their central positions within the interaction network suggest that they may coordinate multiple biological processes potentially influenced by EGCG.

The overall protein–protein interaction network generated using STRING is presented in Figure 5. The results of the degree centrality analysis, identifying the most highly connected proteins, are shown in Figure 6. The distribution of these hub genes within the interaction network is illustrated in Figure 7, where the hub proteins are highlighted in red to emphasise their central positions within the network.

Figure 5. STRING protein–protein interaction network generated from predicted EGCG target genes.

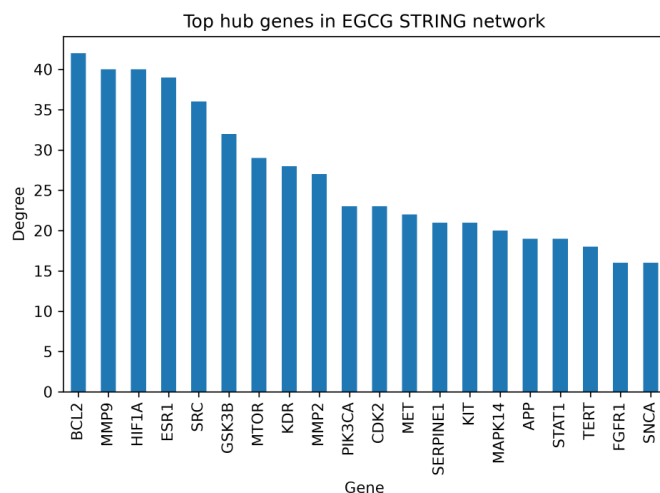


Figure 6. Top hub genes identified in the EGCG protein–protein interaction network based on degree centrality. Higher degree values indicate proteins with a greater number of interactions within the network

EGCG STRING network with hub genes highlighted

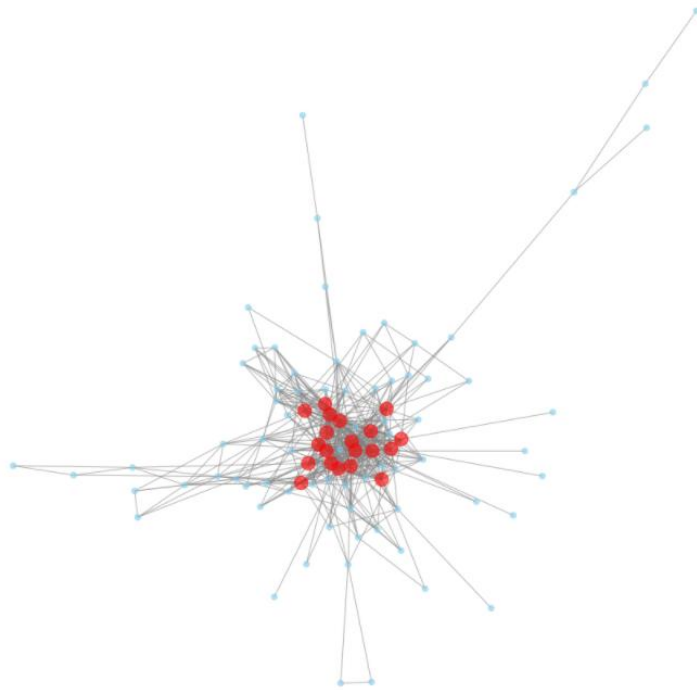
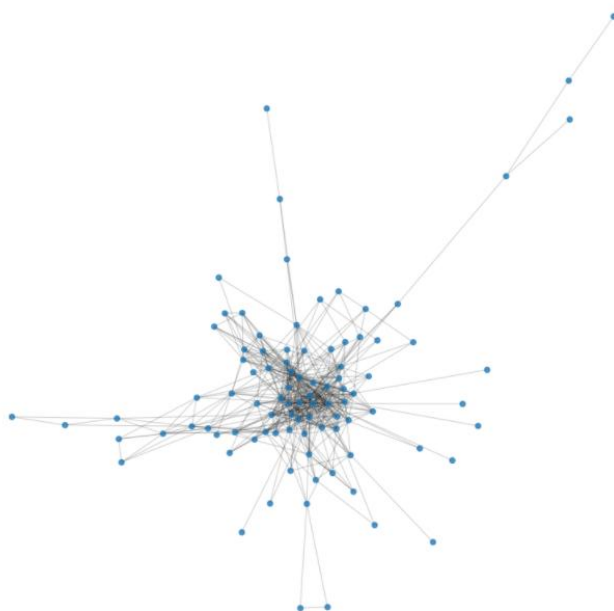


Figure 7. Protein–protein interaction network with hub genes highlighted in red. The highlighted proteins represent the most highly connected nodes identified by degree centrality analysis.

EGCG protein-protein interaction network



4. Discussion

The present study used publicly available bioinformatics resources to investigate the potential molecular mechanisms associated with major bioactive compounds found in matcha, focusing primarily on EGCG. The combination of target prediction, pathway enrichment and protein interaction analysis provides a systems-level overview of biological processes that may be influenced by this compound.

The enrichment analyses consistently highlighted signalling pathways involved in cell survival, proliferation and inflammatory regulation. In particular, the PI3K–Akt signalling pathway appeared in both KEGG and Reactome analyses. This pathway regulates cellular growth, metabolism, apoptosis and responses to oxidative stress. Previous experimental studies have also reported that EGCG can modulate PI3K–Akt activity in various cell types, suggesting that the computational predictions are biologically plausible.

The MAPK signalling pathway was another important finding. MAPK signalling regulates cellular proliferation, differentiation and stress responses and has been implicated in inflammation, cancer biology and cardiovascular disease. The identification of MAPK-related proteins among predicted EGCG targets supports previous experimental observations describing the anti-inflammatory properties of green tea catechins.

Another frequently enriched pathway involved VEGF signalling. VEGF plays an essential role in angiogenesis and vascular homeostasis. The identification of VEGF-related pathways suggests that EGCG may influence mechanisms associated with blood vessel formation and endothelial function. However, these findings represent computational associations and should not be interpreted as evidence of therapeutic effects.

Protein interaction analysis demonstrated that predicted EGCG targets form a densely connected interaction network. Hub proteins such as BCL2, HIF1A, SRC and MTOR are recognised regulators of apoptosis, hypoxia responses and intracellular signalling. Because hub proteins interact with numerous molecular partners, modulation of these proteins could potentially influence multiple downstream biological pathways simultaneously.

One of the strengths of this project is the integration of several independent bioinformatics resources. Combining PubChem, SwissTargetPrediction, Gene Ontology, KEGG, Reactome and STRING provides complementary information that is difficult to obtain from a single

database. The use of reproducible Python notebooks also allows the complete analytical workflow to be repeated or extended with additional compounds in the future.

Nevertheless, several important limitations must be acknowledged. SwissTargetPrediction provides predicted rather than experimentally confirmed protein targets. Similarly, enrichment analysis identifies statistically overrepresented pathways but cannot determine whether those pathways are activated or inhibited under physiological conditions. STRING interaction networks combine evidence from experimental studies, computational predictions and text mining, meaning that not every reported interaction has been experimentally validated.

Furthermore, this study investigated primarily EGCG, whereas matcha contains numerous additional bioactive compounds, including caffeine, L-theanine and other catechins. Future work should compare the predicted molecular targets of these compounds to identify shared pathways, unique molecular effects and potential synergistic interactions.

Finally, the computational results presented here should be regarded as hypothesis-generating rather than definitive biological evidence. Experimental validation using cell culture models, animal studies or clinical investigations would be required before concluding that matcha or EGCG directly modifies the identified signalling pathways or improves human health.

5. Conclusion

This project demonstrated how publicly available bioinformatics resources can be integrated to investigate the potential molecular effects of bioactive dietary compounds.

Target prediction identified approximately 96 unique human protein targets for EGCG. Functional enrichment analysis consistently highlighted signalling pathways associated with PI3K–Akt, MAPK, VEGF, extracellular matrix organisation and protein phosphorylation. Protein–protein interaction analysis identified highly connected hub genes including BCL2, MMP9, HIF1A, ESR1 and MTOR, suggesting that these proteins may occupy central positions within the predicted molecular interaction network.

Although the analyses provide valuable insights into possible biological mechanisms, they remain computational predictions and should not be interpreted as evidence of clinical efficacy. Experimental validation is essential to determine whether the predicted interactions occur

under physiological conditions and whether they contribute to the reported health effects of matcha consumption.

Overall, this project demonstrates the usefulness of systems biology and bioinformatics approaches for generating testable hypotheses regarding the molecular actions of natural products.

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